RIVER VALLEY HIGH SCHOOL
YEAR 6 PRELIMARY EXAMINATION

CANDIDATE NAME													
CENTRE NUMBER	S					CLASS		INDEX NUMBER					

H2 BIOLOGY **9744/04**

Paper 4 Practical **28 Aug 2018**

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: As listed in the Confidential Instructions.

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.
Write your centre number, index number, class and name on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You will lose marks if you do not show your working, or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

Shift	
Laboratory	
For Examiner's Use	
1	**/ 22**
2	**/ 15**
3	**/ 13**
Total	**/ 50**

Answer **all** questions.

1 Lipase, **E**, catalyses the hydrolysis of triglycerides into fatty acids and glycerol.

The substrate for **E** will be the triglycerides present in milk, labelled **M**.

The end-point of this hydrolysis can be determined by using an indicator, **I**, which changes colour when the fatty acids are produced.

You are required to:

- prepare different concentrations of the lipase solution, **E**
- investigate the effect of different concentrations of **E** on the hydrolysis of triglycerides in milk.

You are provided with:

labelled	contents	hazard	volume / cm^3
M	milk	none	40
W	distilled water	none	50
I	indicator solution	stains	30
A	solution of alkali	irritant	40
E	5% lipase solution	irritant	50

You are required to dilute the 5% lipase solution, **E**, to provide a range of known concentrations using **simple** dilution.

Decide on the further concentrations of lipase solution you will use in your investigation in addition to the 5% solution.

You will need to prepare 10 cm^3 of each lipase solution.

(a) **(i)** Prepare the space below to show the concentration of each lipase solution, the volumes of **E** and the volumes of **W**. [3]

Table showing dilution of E

Concentration of lipase solution / %	Volume of E / cm^3	Volume of W / cm^3
1	2.0	8.0
2	4.0	6.0
3	6.0	4.0
4	8.0	2.0
5	10.0	0.0

Shows at least 5 linear dilutions of equal intervals for lipase solution

Correct volumes of E

Correct volumes of W

(ii) Describe and explain the expected trend in the time taken to reach the end-point as the concentration of lipase solution increases. [2]

1. As concentration of lipase increases, time taken to reach end-point decreases
2. More lipase-triglyceride / enzyme-substrate complex formed per unit time
3. More fatty acids formed per unit time

*Before starting the investigation, read through steps **1-8** and prepare a table in **(a)(iii)**.*

Proceed as follows.

1 Prepare **all** the concentrations of lipase solutions you have listed in **(a)(i)**.

2 Put 2 cm^3 of **M** into each test-tube.

3 Put 2 cm^3 of **I** into each test-tube containing **M** and gently shake.

4 Put 3 cm^3 of **A** into each test-tube containing **M** and **I** and gently shake so that all the mixture turns orange. *Note that the mixtures might be different shades of orange.*

5 Put 2 cm^3 of **E** into one of the test-tubes from step **4** and mix well. Wait for 300.0s. This will be the colour of the end-point.

6 Repeat step **5** with **all** concentrations of enzyme solution.

7 Start the stopwatch.

8 Record the time when each end-point is reached. If the time taken to reach end-point for any one concentration is longer than 300.0 seconds, record as ‘more than 300.0’.

(iii) Record your results in a suitable table in the space below. [3]

Table showing effect of lipase concentration on time taken to reach end-point

Concentration of lipase solution / %	**Time taken to reach end-point / s**
1	200.1
2	135.2
3	86.4
4	52.3
5	37.3

Correct heading with units
Whole number for lipase concentration; 1dp for time

Shortest time for 5% lipase and longest time for lowest lipase concentration

(iv) Calculate the rate of hydrolysis for 5% lipase concentration.

You should show your working and use appropriate units. [2]

Rate of hydrolysis for 5% lipase concentration

$= \frac{1}{37.3}$

$= 0.0268\ s^{-1}$

1. Working showing $\frac{1}{\text{time}}$
2. Correct calculation and units

(v) Identify **one** significant source of error in measuring the dependent variable and describe how it affects your results. [2]

1. Visual determination of colour of end-point is subjective
2. May result in over- or under- estimation of time taken to reach end-point

1. Reaction in some test-tubes have started before starting the stopwatch
2. May result in underestimation of time taken to reach end-point

(vi) Other than enzyme concentration, temperature has significant impact on the rate of lipase hydrolysis.

Suggest how you would modify this investigation to obtain an accurate optimum temperature for the activity of **E**. [3]

1. (Equilibrate) **M**, **I**, **A** and **E** in (thermostatically-controlled) water bath set to 10, 20, 30, 40 and 50°C
2. Determine end point using a pH sensor connected to a datalogger
3. Repeat experiment by narrowing intervals for temperatures with shorter time taken to reach end-point
   OR
   Plot of graph of data points and identify the temperature that corresponds to the shortest time taken to reach end point

(b) Some students studied the effect of temperature on the rate of lipase hydrolysis, using a different method that involves determining the presence of triglycerides at various time intervals.

The students' results are shown in Table 1.1.

Table 1.1

Temperature / °C	Rate of lipase hydrolysis / mol dm^{-3} s^{-1}
10	23.0
20	52.0
30	68.5
40	35.0
50	0.5

(i) Describe a chemical test that can be used to determine the presence of triglyceride in a sample solution. [2]

1. Add equal volumes of sample solution and ethanol
2. (Shake vigorously and) centrifuge mixture
3. Decant top ethanol layer into (equal) volume of water
4. If triglyceride were present, white emulsion is observed

(ii) Use the grid to display the results shown in Table 1.1 in an appropriate form. [3]

1. HU: Correct heading with units for both axes
2. P: Accurate plot points
3. C: Smooth and best-fit curve

(iii) Explain the decrease in rate of lipase hydrolysis after 30°C. [2]

1. Increase in kinetic energy
2. breaks hydrogen bonds and hydrophobic interactions in lipase
3. causing active site to lose its three dimension conformation
4. Lipase is denatured

[Total: 22]

2 Tuberculosis (TB) is an infectious disease caused by the bacterium *Mycobacterium tuberculosis. M. tuberculosis* in the body provokes an immune response, resulting in the production of specific antibodies.

In a test used to detect TB, a modified antibody **A** specific to *M. tuberculosis* is used. Binding of this antibody to *M. tuberculosis* gives rise to an observable result when tested with test reagent **X**.

You are provided with:

- the blood serum of three patients in microfuge tubes labelled **P1**, **P2**, and **P3**
- a suspension *of M. tuberculosis* in a microfuge tube labelled **T**
- distilled water in a microfuge tube labelled **W**
- a solution of the modified antibody in a microfuge tube labelled **A**
- test reagent **X** in a microfuge tube labelled **X**

You are recommended to wear suitable eye protection and gloves. Any splashes on skin should be washed off immediately.

You are required to carry out the test and determine the observations associated with a positive test.

You will then test the blood serums and determine which of the patients should be diagnosed with TB.

You should take care when using the Pasteur pipettes to ensure no cross-contamination of samples and reagents occur.

Proceed as follows.

1 Label the wells of the microtiter plate **T**, **W, P1, P2** and **P3**.

2 Add 2 drops of **T** and **W** into the appropriately labelled wells.

3 Add two drops of antibody **A** into wells **T** and **W**.

4 Add two drops of test reagent **X** into wells **T** and **W**, and leave for 10 seconds.

(a) Describe your results of testing samples **T** and **W.** [1]

1. **T** – red colour observed
2. **W** – green colour observed;

5 Add 2 drops of **P1**, **P2** and **P3** into the appropriately labelled wells.

6 Add two drops of antibody **A** into wells **P1**, **P2** and **P3**.

7 Add two drops of test reagent **X** into wells **P1**, **P2** and **P3,** and leave for 10 seconds.

(b) Record your results and conclusions in Table 2.1 [2]

Table 2.1

sample	observation	Presence of *M. tuberculosis*
P1	mixture turned red	present
P2	mixture remained green	absent
P3	mixture turned red	present

1. Correct observations
2. Correct conclusions

(c) Explain why antibody **A** cannot be used to detect if a patient is infected with other species of bacteria. [2]

1. Other species of bacteria will have different antigens / lack the antigens (of *M. tuberculosis*)
2. Antibody **A** cannot bind to the antigen / is specific to *M. tuberculosis* antigen
3. will always show a negative result

(d) Treatment of drug-resistant strains of *M. tuberculosis* requires more than one antibiotic to be used simultaneously. A common treatment involves administering a cocktail solely comprising two component antibiotics, isoniazid and rifampicin.

The relative concentrations of each antibiotic in the cocktail determines its effectiveness. Effective treatment with the cocktail will result in the bacteria being hydrolysed to its constituent biomolecules after 24 hours of exposure.

Table 2.2

cocktail number	**relative concentration of isoniazid to rifampicin**
1	20-80
2	80-20

A student administered two different cocktails, as shown in Table 2.2, on drug-resistant *M. tuberculosis,* before repeating the test in question **2**. He found that both cocktails were ineffective in killing the bacteria.

The student hypothesised that the cocktail is only effective when the relative concentration of each antibiotic component is at least 30%.

Design an experiment to determine the cocktail with the most effective relative concentrations of component antibiotics.

In your plan, you must use:

- a suspension of drug-resistant *M. tuberculosis* in a water-bath at 30°C (3 cm^3 of this culture will be required for inoculation of each additional culture)
- a sterile solution of 100% isoniazid
- a sterile solution of 100% rifampicin
- a water-bath at 30°C
- a colourimeter

You may select from the following sterilised apparatus and plan to use appropriate additional apparatus:

- syringes
- 5 cm^3 microfuge tubes
- timer, e.g. stopwatch
- a biosafety cabinet
- normal laboratory glassware, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes and pipette fillers, glass rods, etc.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary

- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[10]

Mark Scheme

Independent variable:

1. States that independent variable is relative concentration of component antibiotics and uses at least uniformly-spaced relative concentrations

Dependent variable:

2. Intensity of red/green colouration of reaction mixture

Controlled variables:

3. Ref. to controlling temperature at 30°C with thermostatically controlled water-bath

4. Identify and describe another variable to be controlled

Scientific Theory and Reasoning:

5. Explains that hydrolysed bacteria do not have antigens for antibody **A** to bind to

Method:

6. Shows how to perform a simple dilution to obtain the specified relative concentrations

7. Use of colourimeter (to obtain absorbance values)

8. Describe how to determine the cocktail with most effective relative concentrations of antibiotics

Reliability:

9. Performs at least two more repeats and replicates with new reagents

Accuracy:

10. Repeating with decreased intervals of relative concentrations to obtain more data to achieve experimental aim

Control:

11. Perform a negative control with cocktail replaced with equivolume of distilled water

Recording:

12. Shows how results are to be presented in the form of a table with IV and DV in appropriate column/rows

Risk/safety:

13. Refers to use of a biosafety cabinet in performing the procedure / Test reagents (**X**, **A**) are irritants, wear goggles and gloves to prevent contact / Antibiotics may cause allergic reactions, wear goggles and gloves to prevent contact

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SAMPLE REPORT

Aim	To investigate the most effective relative concentrations of isoniazid to rifampicin to treat tuberculosis
Independent variable	Relative concentrations of isoniazid to rifampicin - 70-30, 60-40, 50-50, 40-60, 30-70
Dependent variable	Intensity of green colouration of reaction mixture
Controlled variable	1. Temperature of thermostatically-controlled water-bath – 30°C
	2. Volume of drug administered – 0.5 cm^3
	3. Volume of drug-resistant *M. tuberculosis* culture – 3 cm^3
	4. Duration of incubation – 24 hours

Scientific Theory and Reasoning

Effective treatment with the antibiotic cocktail results in the *M. tuberculosis* being hydrolysed to its constituent biomolecules. The hydrolysed bacteria do not have antigens for antibody **A** to bind to, and when tested with antibody **A** and test reagent **X**, the reaction mixture will remain green.

1. Perform a simple dilution to obtain different relative concentrations of antibiotics to the dilution table shown below.

Table showing dilution of component antibiotics

Drug	Relative concentrations of isoniazid to rifampicin	Volume of drug prepared / cm^3	Volume of 100% isoniazid added / cm^3	Volume of 100% rifampicin added / cm^3
3	70-30	5.0	3.5	1.5
4	60-40	5.0	3.0	2.0
5	50-50	5.0	2.5	2.5
6	40-60	5.0	2.0	3.0
7	30-70	5.0	1.5	3.5

2. Perform all experiments in a biosafety cabinet.
3. To a microfuge tube, add 1 cm^3 of drug D1. Label this tube “Drug D1”.

4. Repeat step 3, replacing the 1 cm^3 of drug D1 with the drugs containing different relative concentrations of isoniazid and rifampicin. Label the tubes accordingly.
5. Add 3 cm^3 of the drug-resistant *M. tuberculosis* into each of the tubes.
6. Incubate the bacteria in a thermostatically-controlled water bath set at 30°C for 24 hours.
7. After incubation, label the wells of a microtiter plate. Test sample from drug D1 with antibody **A** and test reagent **X**.
8. After 1 minute, place 1 cm^3 of reaction mixture into a cuvette, and measure the intensity of green colouration with the colourimeter. Record the absorbance values
9. Repeat steps 7-8 for the remaining samples tested.
10. Performs two replicates and repeat the experiment twice with new reagents.
11. After determining the relative concentration of component antibiotics that is most effective in killing *M. tuberculosis*, decrease the intervals of relative concentrations tested for more accurate determination of the relative concentration of component antibiotics needed to achieve the experimental aim.
12. The cocktail with the most effective relative concentration of antibiotics will result in a reaction mixture with the highest intensity of green colouration.

Table showing the effect of relative antibiotic concentrations on absorbance of reaction mixture

Relative concentrations of isoniazid to rifampicin	Absorbance, A			
	A_1	A_2	A_3	$\bar{A}$
70-30				
60-40				
50-50				
40-60				
30-70				

Risk Assessment:

1. *M. tuberculosis* is infectious. Perform experiment in a biosafety cabinet.
2. *M. tuberculosis* is infectious. Wear gloves and goggles to avoid contact with the bacterium.
3. Isoniazid and rifampicin may trigger allergic reactions. Wear gloves and goggles to avoid contact with the antibiotics.
4. Test reagent **X** is an irritant. Wear gloves and goggles to avoid contact with the test reagent.

[Total: 15]

3 During this question you will require access to a microscope and slide **S1**.

You are required to use a sharp pencil for drawings.

A blood smear can be used to look for abnormalities in blood cells. Observations made from blood smears allow doctors to diagnose certain blood disorders or other medical conditions.

In a blood smear, mature red blood cells and two categories of white blood cells can be observed clearly. The two categories of white blood cells are lymphocytes and phagocytes.

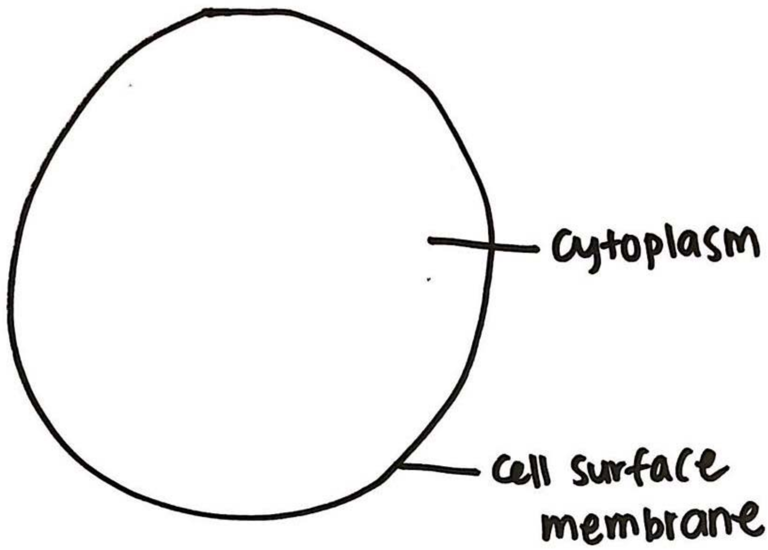
Mature red blood cells do not have nucleus, but white blood cells contain a nucleus. The shape of lymphocyte's nucleus is round but the shape of phagocyte's nucleus is lobed (dumbbell-shaped, C-shaped etc.).

(a) Observe the cells in slide **S1**.

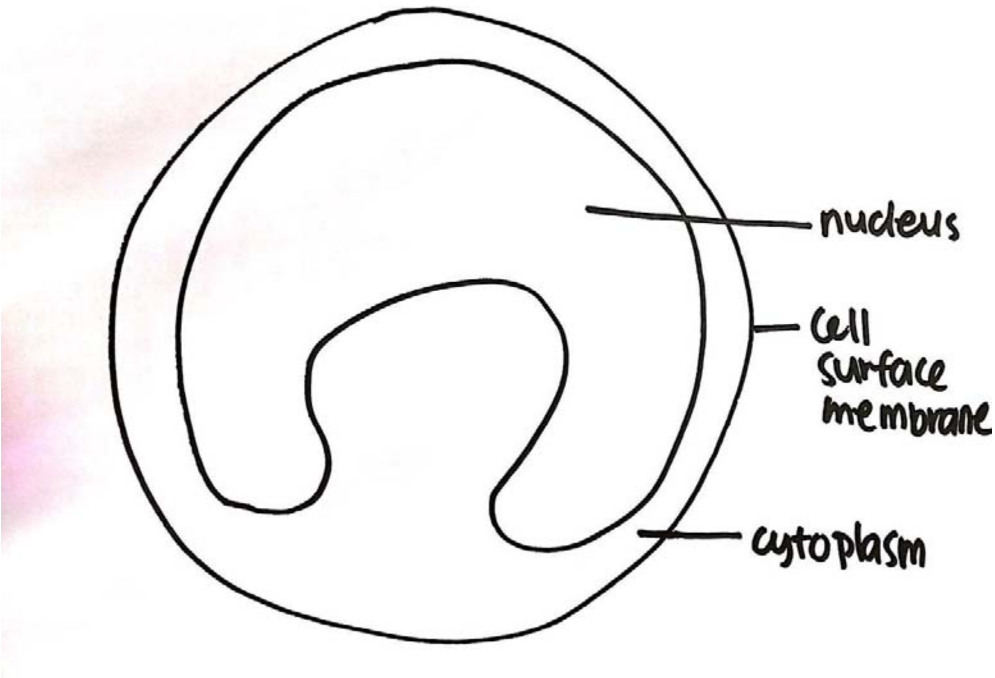
Identify **one** red blood cell and **one** phagocyte.

Use the space provided to draw to the same scale, labelled diagrams of [5]

(i) a red blood cell,



(ii) a phagocyte.



Correct cells drawn – RBC – cell without nucleus

Correct cells drawn – Phagocyte – cell with nucleus

Proportions – Red blood cell smaller than phagocyte

Two correct labels – Cell surface membrane, cytoplasm, nucleus

Continuous clear lines

Slide **S1** is a microscope slide with blood smear of individual **A**.

Fig 3.1 is a photomicrograph of a blood smear of individual **B** viewed at x400.

Both blood smears have been stained using the same technique.

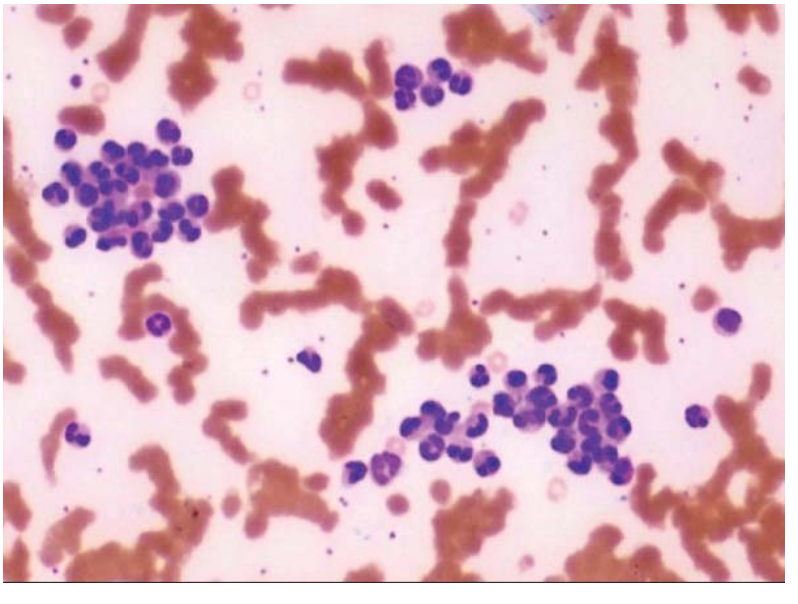


Fig 3.1

(b) **(i)** Using a suitable form, record observable differences between the blood smear of individual **B** in Fig 3.1 and the individual **A** on slide **S1**. [3]

Table showing the observable difference between Fig 3.2 and S1

Feature	Fig 3.2	S1
Number of WBC/cells with nucleus	More WBC	Lesser WBC
Clustering of white blood cell	Yes	No
Clustering of red blood cell	Yes	No

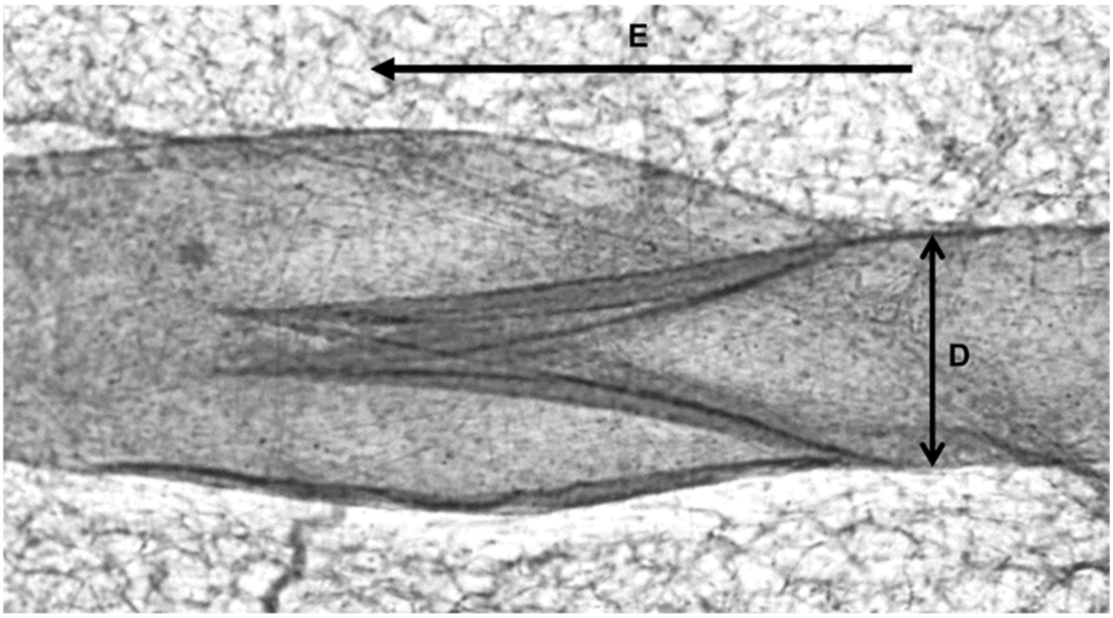
(ii) Individual **B** suffers from pain and shortness of breath. With reference to Fig 3.1, and your own knowledge, suggest reasons for these symptoms. [2]

1. **Red blood cell obstruct oxygen transport leading to organ damage**
2. **Red blood cells accumulate in the spleen (for destruction), leading to enlargement of spleen**
3. **Red blood cell haemolyse easily, resulting in anaemia**
4. **Possible inflammation event where cytokines are release, leading to pain**

Lymph is a fluid containing white blood cells, especially lymphocytes, the cells that attack bacteria in the blood. Lymph flows through a network of lymphatic vessels, transporting white blood cells to tissues of the lymphatic system. The lymphatic system includes the bone marrow, thymus and lymph nodes.

Fig 3.2 shows the longitudinal section of a lymphatic vessel with a pair of flap-like structures, known as a valve. Arrow **E** in the photomicrograph shows the direction of lymph flow.

You are not expected to be familiar with this specimen.



X 400.0

Fig 3.2

(c) Calculate the actual diameter of the narrowest region of the lymph vessel, indicated by line **D**. Show your working clearly. [2]

Actual length = Image size/magnification

Actual length = [Image size (measure after printing)/400.0] x 10^4

77.5 (Image size 3.1) or 75 (Image size 3.0)

Actual diameter : _____________ µm

(d) With reference to Fig 3.2, describe an observable structural adaptation that allows the lymphatic vessel to transport lymph around the body. [1]

1. **Has a lumen which can be filled lymph**
2. **Has valves/muscular flaps which does not allow blood to flow in the opposite direction/blackflow of blood**

[Total: 13]

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